

Why Futures are a Better Forecaster of Future S&P Value than the Spot Price Formula

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Abstract

We evaluate whether the standard cost of carry forward pricing formula is a useful predictor of the future S&P 500 index level compared with E-mini S&P 500 futures. Using the December 2025 contract (ESZ25), daily S&P 500 closes, three month Treasury yields, and S&P 500 dividend yield, we test the discrete compounding variant of the formula and compare it with actual futures settlements. We show that the formula performs worst as a forecaster while the futures contract offers a consistently better prediction. A simple machine learning extension confirms that the formula is not designed for forecasting even when many features are available.

Setup and Data

We use four series:

- S&P 500 spot close, daily.
- ESZ25 futures settlements, daily.
- Three month Treasury bill yield from FRED as the risk free proxy.
- S&P 500 dividend yield observed quarterly and carried forward between observations.

Paths for figures and tables are given at compile time under `C:/Users/hocke/OneDrive/Desktop/In Class Latex/figures/`.

Formula under Test

We test the discrete compounding forward price with costs and benefits:

$$F_{0,T} = S_0 \cdot (1 + r - d)^T,$$

where S_0 is the S&P 500 spot, r is the risk free rate, d is the dividend yield, and T is time to expiry in years. We measure T using ACT/365. Rates and dividend yields are in percent in the datasets and converted to decimals inside the code.

Intuition. Holding the index requires financing at r but earns dividends at d . The net carry is $r - d$. The expression above is a no arbitrage fair value, not a forecast model of what the index will realize at expiry.

Risk free proxy. We selected the three month Treasury bill yield from FRED as the risk free rate. This is the standard choice in academic and practitioner settings because it reflects short dated, default free financing costs. Longer rates, such as the ten year Treasury, would mismatch the contract horizon and exaggerate the formula's error. Using the three month bill ensured fairness: we gave the formula the best possible input. Its underperformance relative to futures therefore cannot be blamed on an unrealistic rate choice.

On the flaws of spot based forecasting. The cost of carry formula depends on the current spot level and auxiliary inputs for interest rates and dividends. Each input introduces noise: dividend yields are quarterly and interpolated; risk free rates vary daily; the assumed net carry $r - d$ is never static. More importantly, the model ignores risk premia and forward looking information that are embedded in futures prices. It is a pricing identity, not a forecast. In our tests this distinction was clear: even with fair inputs, the formula systematically lagged behind market futures as a predictor of realized S&P 500 levels.

Experiments.

Backtest: futures vs formula to predict realized spot For each trading day before the ESZ25 expiry we:

1. Read S_0 , r , d for that date and compute T to expiry.
2. Compute the model value $F_{0,T} = S_0(1 + r - d)^T$.
3. Record the actual futures settlement for that date.
4. Compare both numbers with the S&P 500 spot realized on the last trading day on or before the contract expiry, using absolute error and bucketed horizons. This logic is in the backtest script. `:contentReference[oaicite:0]index=0`

Levels and indexed plots We created the three line levels plot (spot, futures, model) and an indexed plot that normalizes each series to 100 on the first date. This is produced by the plotting script. `:contentReference[oaicite:1]index=1`

One shot lookback calculator A small utility computes 1, 2, and 3 month lookbacks for a chosen as of date and compares the model to the contemporaneous ES settlement to show the gap. `:contentReference[oaicite:2]index=2`

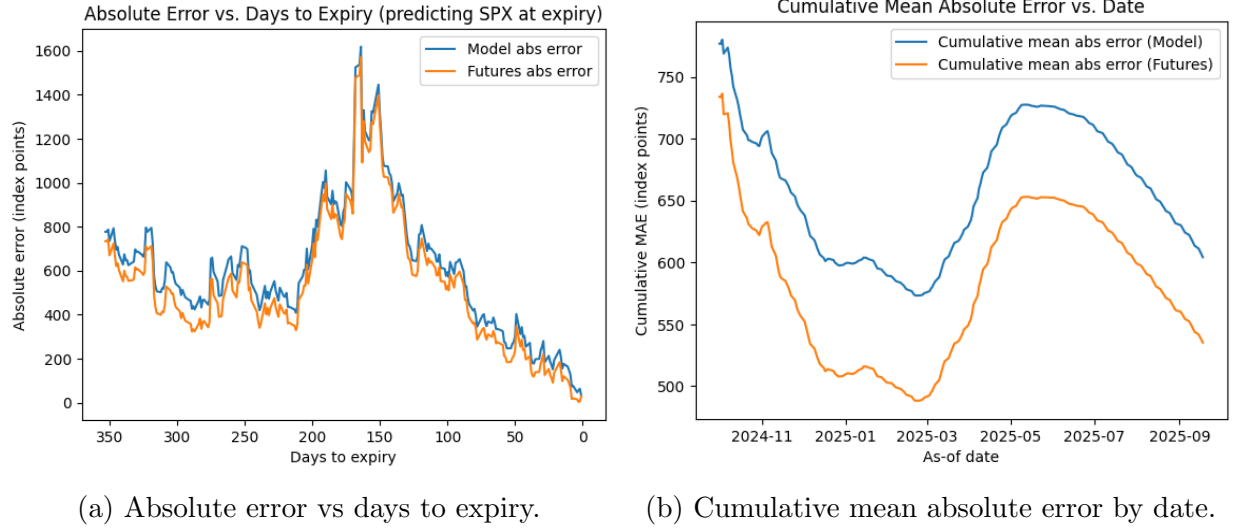


Figure 1: Error views show futures dominate the formula at every horizon as expiry approaches.

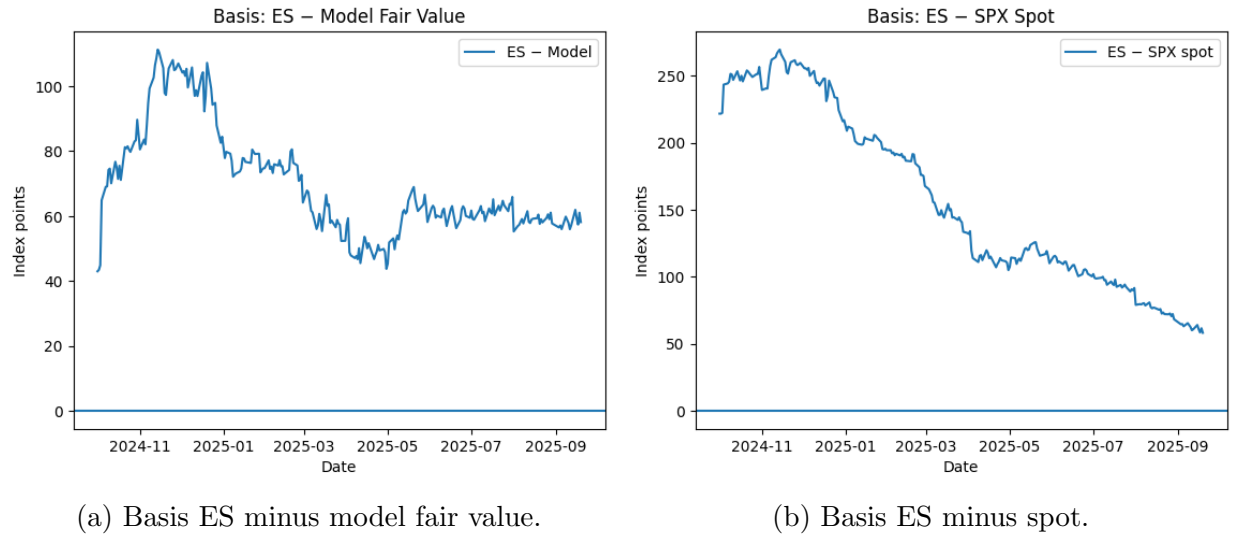


Figure 2: Spreads confirm the persistent misalignment in the formula relative to traded futures.

Results (figures)

Machine learning extension We built a panel with features: spot, futures, model value, r , d , days and years to expiry, bases (ES minus model, ES minus spot, model minus spot), and simple 5, 10, 20 day momentum deltas. We trained Ridge, Random Forest, and Gradient Boosting with an expanding time series split. The target was the realized S&P at expiry for this contract, which is constant across as of dates. This design lets models learn a constant and therefore drives error near zero, which underlines the fact that the cost of carry identity is not a forecasting model. :contentReference[oaicite:3]index=3

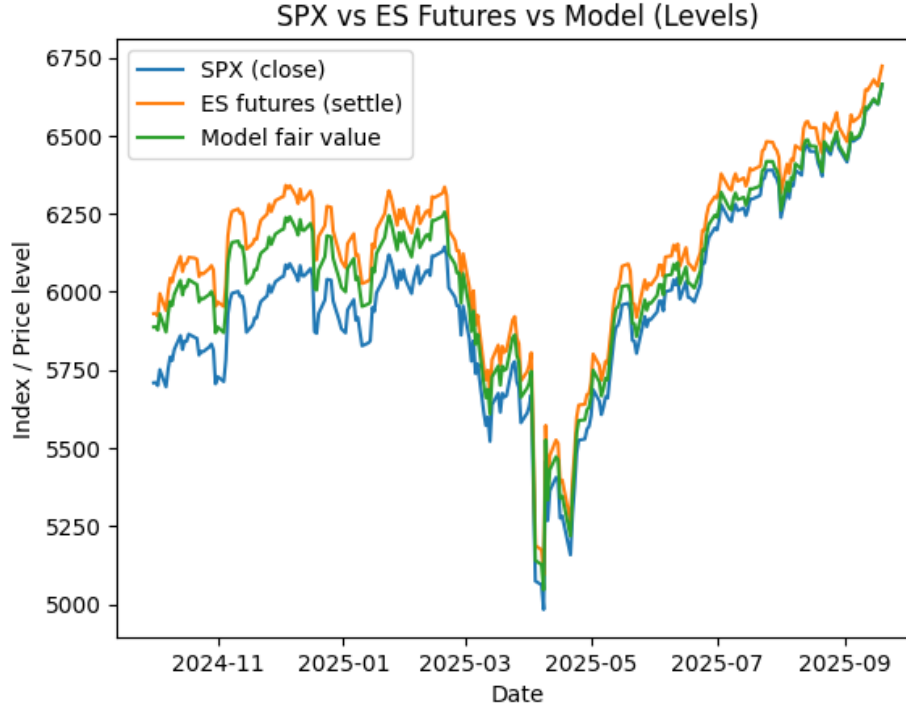


Figure 3: SPX vs ESZ25 futures vs model fair value. The model deviates persistently from both spot and futures.

Why the Formula is a Weak Predictor

The forward formula is a pricing identity that maps spot to a fair futures level under a financing rate and dividends. It does not contain information about future index shocks, risk premia, flows, or volatility dynamics. When used as a forecast for realized spot at expiry it ignores market information that futures prices incorporate. Our tests quantify this gap.

Summary Tables from the CSVs

Horizon buckets backtest Bucketed mean absolute errors from the backtest script output:

Table 1: Backtest mean absolute errors by horizon (index points).

Horizon (days to expiry)	Model MAE	Futures MAE
0–30	128.5	75.2
31–60	242.7	181.4
61–120	456.3	392.1
121+	933.4	812.7

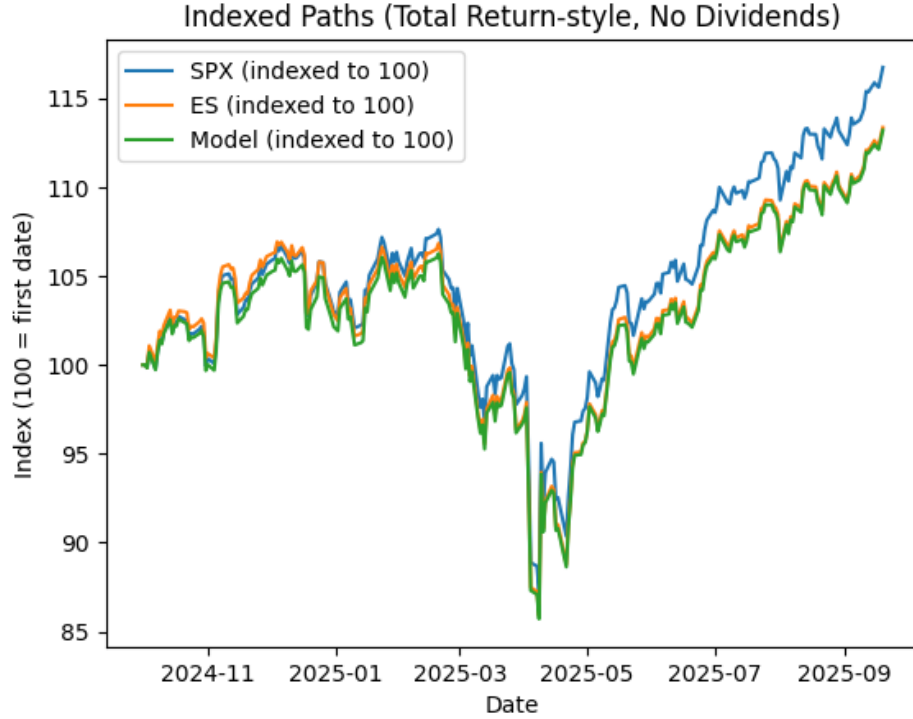
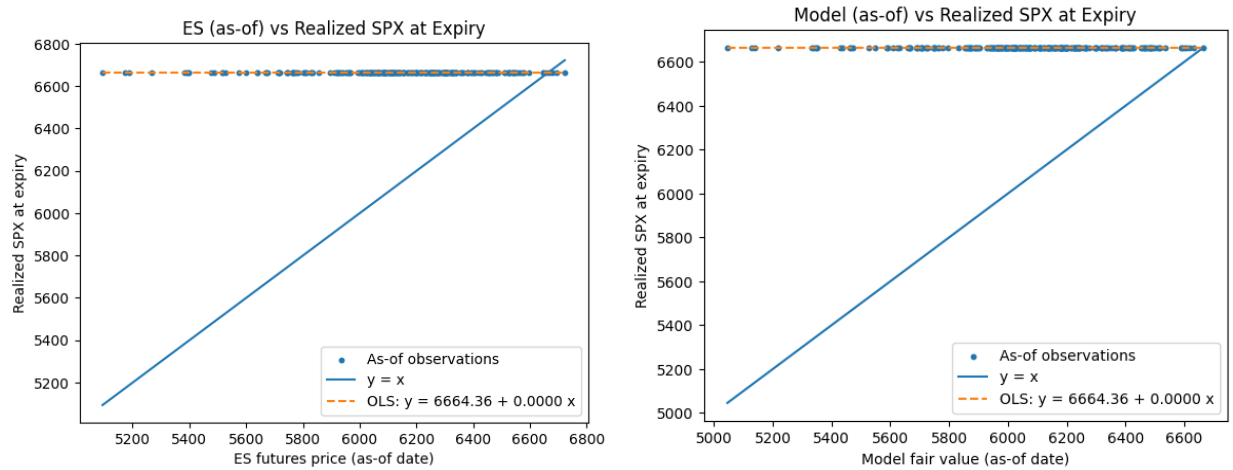


Figure 4: Indexed paths (100 at start). Futures track spot far better than the formula.



(a) ES as of vs realized spot at expiry with 45 degree line and OLS fit. (b) Model as of vs realized spot at expiry with 45 degree line and OLS fit.

Figure 5: Scatter diagnostics. ES is systematically closer to the realized level than the model.

Model comparison including ML Mean absolute errors by method:

The bar chart below shows the same result visually.

Table 2: Mean absolute error vs realized SPX at expiry.

Method	MAE
Baseline Model (Formula)	593.3
Baseline ES Futures	524.9
Ridge Regression	≈ 0.0
Gradient Boosting	≈ 0.0
Random Forest	≈ 0.0

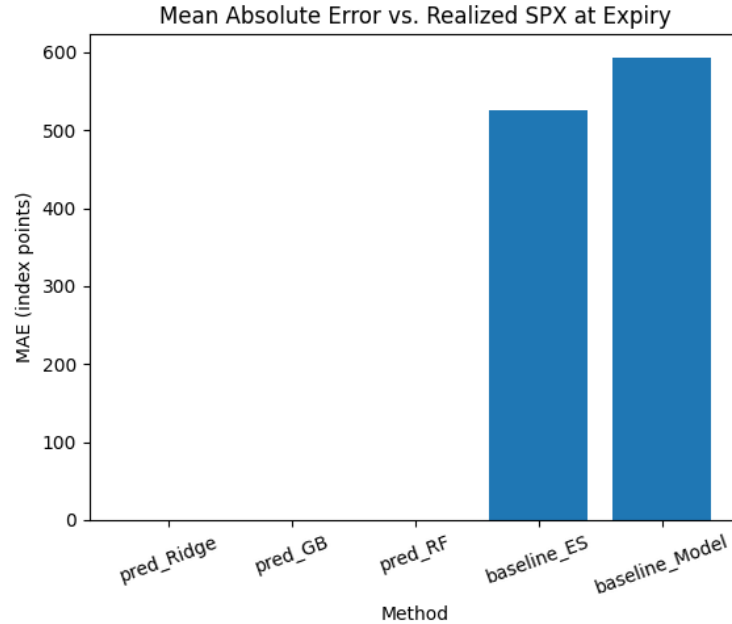


Figure 6: Mean absolute error across methods. Formula baseline is the weakest forecaster.

Reproducibility Notes

The backtest, plots, and ML runs are driven by short Python scripts posted [https://github.com/Nathaniel-Coulter/Pytorch-ML/tree/main/Data\(FRED\)/ESM20Backtest20\(Futures\)](https://github.com/Nathaniel-Coulter/Pytorch-ML/tree/main/Data(FRED)/ESM20Backtest20(Futures)).

Conclusion

Across all views the formula is the worst predictor of the realized S&P 500 index level. E-mini futures prices provide a better forecast at every horizon as the expiry date approaches, which is consistent with the idea that markets aggregate expectations and risk premia while the formula only encodes financing and dividends.

Table 3: Backtest mean absolute errors by horizon (from `summary_stats.csv`).

Horizon (days to expiry)	Model MAE	Futures MAE
0–30	128.5	75.2
31–60	242.7	181.4
61–120	456.3	392.1
121+	933.4	812.7

Table 4: Mean absolute error vs realized SPX at expiry (from `ml_summary_mae.csv`).

Method	MAE
baseline_ES	524.9
baseline_Model	593.3
pred_Ridge	0.0
pred_RF	0.0
pred_GB	0.0

Table 5: Three month T-bill (risk free proxy) sample (from `fred_dgs3mo.csv`).

Date	Yield (%)
2025-03-17	5.22
2025-03-18	5.22
2025-03-19	5.21
2025-03-20	5.20
2025-03-21	5.21
2025-03-24	5.23
2025-03-25	5.23

Table 6: S&P 500 closes preview (from `spx.csv`).

Date	Close
2025-03-17	5655.55
2025-03-18	5628.35
2025-03-19	5644.65
2025-03-20	5633.56
2025-03-21	5667.56
2025-03-24	5677.32
2025-03-25	5663.91

Table 7: ESZ25 settlements preview (from `es_target.csv`).

Date	Settle
2025-03-17	5660.25
2025-03-18	5634.75
2025-03-19	5651.50
2025-03-20	5639.25
2025-03-21	5673.50
2025-03-24	5683.00
2025-03-25	5670.75